Semantic Correspondence with Visual Foundation Models

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Abstract

Semantic correspondence aims to establish pixel-level matches between semantically similar object parts across different images, under strong variations in viewpoint, scale, and appearance. This work evaluates visual foundation models for semantic correspondence on SPair-71k, focusing on DINOv2, DINOv3, and SAM. We compare frozen feature matching, light fine-tuning of the last backbone layers, and Low-Rank Adaptation (LoRA), and we evaluate window soft-argmax as a refinement of the final correspondence prediction. Our results show that self-supervised DINO features are better suited than SAM encoder features for keypoint-level matching under this protocol. DINOv2 provides the strongest zero-shot baseline, while DINOv3 achieves the best overall performance after light fine-tuning. LoRA offers a parameter-efficient adaptation strategy, but remains below direct last-layer fine-tuning in our experiments. Finally, window soft-argmax provides small but consistent localization gains, especially at stricter PCK thresholds. The code is available in the repository at `https://github.com/PasqualeSarcina/AML-polito`*, where additional experiments and the complete set of results are also provided.*

1. Introduction

Semantic correspondence consists in finding pixel-level matches between semantically related object parts across different images, for example matching the eye of an animal in one image to the corresponding eye in another. This task is challenging because object instances may differ in viewpoint, scale, pose, background, and appearance, and because models must distinguish visually similar but semantically different parts.

Recent visual foundation models provide a strong starting point for this problem. Self-supervised Vision Transformers such as DINOv2 and DINov3 learn robust dense representations that transfer well to pixel-level tasks [6], while SAM provides powerful segmentation-oriented features. However, these models are trained for general visual representation learning or segmentation, not explicitly for semantic correspondence. As a result, their frozen features may not be optimal for precise part-level matching, especially on challenging benchmarks such as SPair-71k [3].

This work studies how to adapt such models to semantic correspondence without resorting to full fine-tuning. We compare a training-free baseline, light fine-tuning of the last Transformer layers, and Low-Rank Adaptation (LoRA), which introduces trainable low-rank updates while keeping most pretrained weights frozen [5]. We also evaluate window soft-argmax as a localization refinement over hard argmax, following the idea that local score distributions can provide more precise predictions than a single discrete maximum [4].

Our experiments show that light fine-tuning is the most effective adaptation strategy. DINOv3 achieves the best overall results, while DINOv2 remains a strong zero-shot baseline. SAM improves after adaptation but remains less suitable for fine-grained keypoint correspondence. Finally, LoRA provides a parameter-efficient alternative and improves all backbones, but does not match the best light fine-tuning results. In addition, motivated by recent work on diffusion-based correspondence [7, 8], we also consider Stable Diffusion features as a complementary source of spatially structured representations.

2. Background

Vision Foundation Models. We rely on large-scale pretrained visual backbones to extract dense image representations. In particular, we evaluate DINOv2 ViT-B/14, DINOv3 ViT-B/16, and SAM ViT-B/16. DINOv2 and DINOv3 are self-supervised Vision Transformers designed to learn transferable visual features, while SAM is primarily optimized for promptable segmentation. This distinction is relevant for semantic correspondence, since the task requires not only object-level understanding, but also fine-grained localization of corresponding parts.

Semantic Correspondence Benchmark. We evaluate all methods on SPair-71k, a large-scale benchmark for semantic correspondence containing 70,958 image pairs across 18 object categories [3]. The dataset provides keypoint annotations under strong variations in viewpoint, scale, occlusion, and truncation, making it suitable for evaluating part-level matching between different object instances. Compared to earlier benchmarks such as PF-PASCAL and PF-WILLOW, SPair-71k offers a larger and more challenging evaluation setting.

Parameter-Efficient Fine-Tuning. Since full fine-tuning of large pretrained backbones can be costly and may lead to overfitting, we compare light fine-tuning of the last layers with Low-Rank Adaptation (LoRA). LoRA freezes the pretrained weights and introduces trainable low-rank updates inside selected layers, reducing the number of trainable parameters while still allowing task-specific adaptation [5].

Correspondence Prediction. Dense correspondences are obtained from feature similarity maps. A standard hard argmax selects the target location with maximum similarity, but this prediction is constrained by the discrete patch grid. To reduce localization errors, we also use window soft-argmax, which refines the prediction by computing a local weighted average around the maximum response. This allows a more precise estimate of the target point and is particularly useful when the correct correspondence lies between neighboring patches [4].

3. Methodology

3.1. Training-Free Semantic Correspondence

We first evaluate the zero-shot semantic correspondence capabilities of DINOv2, DINOv3 and Segment-Anything. We utilize the SPair-71k benchmark [3], a comprehensive dataset comprising 70,958 image pairs that exhibit extreme intra-class variations in viewpoint, scale, and occlusion. Our evaluation focuses on the standard test split of 12,234 pairs, where the objective is to predict the corresponding target location for each annotated source keypoint across varying object instances.

Given a source image I_s and a target image I_t, we extract features using frozen ViT-B backbones. To align with the native architectural stride of each model, images are resized to 518×518 for DINOv2, 512×512 for DINOv3, and 1024×1024 for SAM; keypoints and target bounding boxes are scaled consistently to the same resized image space.For DINOv2 and DINOv3, we discard non-spatial tokens such as the global [CLS] token and use only spatial patch tokens for matching. Instead, we extract the post-Layer Normalization spatial patch tokens from the final layer. For SAM, we extract the dense image-encoder feature map before the prompt encoder and mask decoder, since our protocol uses SAM only as a frozen visual feature extractor. Instead, we extract the dense feature maps from the encoder's neck, which provides a spatial feature map originally designed for promptable segmentation.

With a given patch size $p \in \{14, 16\}$, the output feature map natively forms a spatial grid of dimensions $h \times w$, where $h = H/p$ and $w = W/p$. Consequently, this spatial discretization yields a 37×37 grid for DINOv2, a 32×32 grid for DINOv3 and a 64×64 grid for SAM. For both DINO architectures, each spatial element is a D-dimensional vector ($D = 768$), culminating in a dense feature tensor $F \in \mathbb{R}^{h \times w \times D}$. Conversely, due to its larger 1024×1024 input resolution and $p = 16$, SAM produces a higher-resolution 64×64 feature grid, although this does not necessarily imply better keypoint-level semantic matching. Furthermore, while its internal ViT backbone maintains a hidden dimension of 768, SAM's neck architecture projects these embeddings into a lower-dimensional subspace ($D = 256$), yielding a final dense feature tensor $F \in \mathbb{R}^{h \times w \times D}$. To extract the descriptor for a source keypoint at continuous pixel coordinates (x_s, y_s), we map it to the discrete grid indices $(i_s, j_s) = (\lfloor y_s/p \rfloor, \lfloor x_s/p \rfloor)$ to retrieve the corresponding patch token $f_s \in \mathbb{R}^D$.

We then compute the cosine similarity between f_s and all spatial descriptors in the target feature map $F_t \in \mathbb{R}^{h \times w \times D}$. The matching target patch (i_t, j_t) is identified via a spatial argmax operation over the 2D similarity map:

$$(i_t, j_t) = \arg\max_{i,j} \frac{f_s \cdot F_t(i,j)}{\|f_s\| \|F_t(i,j)\|} \tag{1}$$

The predicted target coordinate is then obtained in the resized image space by assigning the match to the geometric center of the selected target patch: $(\hat{x}_t, \hat{y}_t) = \left(j_t \cdot p + \frac{p}{2},\ i_t \cdot p + \frac{p}{2}\right)$. Since keypoints and bounding boxes are scaled consistently to the resized image resolution, PCK is computed in the same coordinate system: $(\hat{x}_t, \hat{y}_t) = (j_t \cdot p + p/2, i_t \cdot p + p/2)$.

3.1.1. Evaluation Metric: Percentage of Correct Keypoints

To evaluate the accuracy of the predicted spatial correspondences, we adopt the Percentage of Correct Keypoints (PCK) metric, following standard protocols established in recent dense matching literature [7]. Let $\hat{p}_t = (\hat{x}_t, \hat{y}_t)$ denote the predicted target keypoint mapped back to the continuous image domain, and let $p_t^* = (x_t^*, y_t^*)$ represent the corresponding ground truth coordinates. The spatial prediction error is quantified via the Euclidean distance between these two points, $\|\hat{p}_t - p_t^*\|_2$. To account for object-scale variations, the correctness threshold is normalized by the size of the target object bounding box. Specifically, the acceptable error radius is proportional to the maximum dimension of the target object's bounding box.

Formally, a predicted keypoint k is considered a correct match if it satisfies the condition:

$$\|\hat{p}_{t,k} - p_{t,k}^*\|_2 \le \alpha \cdot \max(H_{\text{bbox}}, W_{\text{bbox}}) \tag{2}$$

where H_{bbox} and W_{bbox} denote the height and width of the ground truth bounding box in the target image, and α is a predefined tolerance factor. To comprehensively analyze the model's localization precision across varying degrees of strictness, we report the overall PCK at multiple thresholds, specifically $\alpha \in \{0.05, 0.10, 0.20\}$.

3.2. Light Fine-Tuning of the Last Layers

To improve the semantic correspondence capabilities of the foundational baselines, we introduce a parameter-efficient supervised fine-tuning stage while maintaining the core feature extraction pipeline. We investigate three adaptation depths by unfreezing the last one, two, or three Transformer layers. This allows us to study how much task-specific adaptation can be introduced before degrading the pretrained representation. This layer-wise analysis is designed to evaluate whether limited backbone adaptation improves keypoint-level matching.

A relevant design choice in our fine-tuning pipeline is the input spatial resolution. Since semantic correspondence requires localized patch-level descriptors, we keep the high-resolution inputs used in the training-free baseline: 518×518 for DINOv2, 512×512 for DINOv3, and 1024×1024 for SAM. This choice is consistent with prior evidence that higher-resolution inputs are beneficial for dense downstream tasks, where small object parts may be poorly represented at lower resolutions [6].

To optimize the networks for semantic correspondence, we use a supervised adaptation of the Information Noise-Contrastive Estimation (InfoNCE) objective [2], inspired by dense spatial formulations such as DenseCL [1]. For a given pair of source and target images, the loss is computed independently for each of the K annotated keypoints. First, we extract a D-dimensional continuous feature descriptor $q \in \mathbb{R}^D$ from the source feature map at the exact sub-pixel coordinate of the keypoint, utilizing bilinear interpolation. This vector serves as the "query" anchor.

We compare this query against the dense grid of target patch descriptors $k_i \in \mathbb{R}^D$, yielding a spatial token grid of dimensions $h \times w$ (i.e., 37×37 for DINOv2, 32×32 for DINOv3 and 64×64 for SAM). The matching task is thus formulated as a dense multi-class classification problem over this grid. By designating the descriptor at the ground-truth target patch as the positive key k^+ and the remaining $(h \times w) - 1$ descriptors as negative noise samples, the dense InfoNCE loss is defined as:

$$\mathcal{L}_{\text{InfoNCE}} = -\log\left(\frac{\exp(q \cdot k^+/\tau)}{\sum_{i=1}^{h\times w}\exp(q \cdot k_i/\tau)}\right) \tag{3}$$

where the dot product represents cosine similarity between L_2-normalized vectors, and $\tau = 0.07$ acts as a temperature scaling parameter. Minimized via cross-entropy, this objective forces the network to reshape its latent space—maximizing the mutual information between semantically paired regions while effectively repelling non-matching local spatial features.

3.3. Correspondence Refinement with Window Soft-Argmax

In the baseline prediction rule described above, final correspondences are obtained by applying hard argmax over the target similarity map. However, this rule has two limitations: it restricts predictions to discrete patch locations and ignores the local structure of the similarity response around the maximum. In practice, hard argmax assigns the prediction to the center of the selected feature patch, e.g. using a patch stride of 14 pixels for DINOv2 and 16 pixels for DINOv3 and SAM. Consequently, if a true correspondence physically lies on the boundary between adjacent patches, this operation introduces an unavoidable quantization error. This effect is most relevant at stricter PCK thresholds, where small localization errors can change whether a match is counted as correct.To alleviate these quantization artifacts, we replace the standard operator with the Window Soft-Argmax formulation proposed by Zhang et al. [4]. Let p_0 denote the discrete 2D grid coordinate identified by the initial hard argmax over the spatial similarity map S. Rather than accepting p_0 as the final prediction, we extract a local $w \times w$ neighborhood window, denoted as $\mathcal{N}(p_0)$, centered strictly around this peak. We first compute a localized spatial probability distribution $\mathbf{P}(p)$ by applying a temperature-scaled softmax function over the similarity scores within this window:

$$\mathbf{P}(p) = \frac{\exp(S(p)/\tau)}{\sum_{q\in\mathcal{N}(p_0)}\exp(S(q)/\tau)} \tag{4}$$

where τ is a temperature hyperparameter that controls the sharpness of the distribution. A properly tuned τ prevents the probability mass from collapsing into a single peak (which would revert the operation to a hard argmax) or spreading too uniformly across the window. The final continuous sub-pixel coordinate $\hat{p}$ is subsequently calculated as the spatial expectation—the weighted sum of all discrete coordinates p within the window, scaled by their corresponding softmax probabilities:

$$\hat{p} = \sum_{p\in\mathcal{N}(p_0)} p \cdot \mathbf{P}(p) \tag{5}$$

By formulating the prediction as a localized expected value, the model effectively interpolates the target position at a sub-pixel level, seamlessly recovering the fine geometric details lost during the discrete patch embedding process. In our experiments, we define the local neighborhood using a window radius of $r = 3$, resulting in a 7×7 window ($w = 7$) and τ to 0.05.

3.4. Parameter-Efficient Adaptation via LoRA

To further reduce the number of trainable parameters compared to direct last-layer fine-tuning, we investigate Low-Rank Adaptation (LoRA) [5] as a parameter-efficient adaptation strategy. Instead of directly updating the pretrained backbone weights, LoRA keeps them frozen and adds trainable low-rank update matrices to selected self-attention projections.

Formally, for a pre-trained weight matrix $W_0 \in \mathbb{R}^{d\times d}$ within the attention mechanism, the learned update is re-parameterized as:

$$W = W_0 + \Delta W = W_0 + BA \tag{6}$$

where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times d}$ are low-rank trainable matrices, constrained such that the rank $r \ll d$. In our implementation, we specifically target the query, key, and value projection matrices (W_q, W_k, W_v) within the Transformer layers. These low-rank adapters are optimized end-to-end using the same keypoint-level dense InfoNCE objective defined in Eq. 3.

During inference, the learned LoRA modules operate in tandem with the frozen backbone. Dense correspondences are established via the adapted cosine similarity maps and subsequently localized at the sub-pixel level utilizing the Window Soft-Argmax strategy. This modular approach seeks an optimal balance between computational efficiency and geometric precision, enabling the model to adapt to the complex intra-class variations of the SPair-71k dataset utilizing only a fraction of the trainable parameters required for standard layer fine-tuning.

3.5. Extension: Cross-Dataset Generalization

To assess the generalization capability of the fine-tuned models, semantic correspondence has also been evaluated on additional datasets beyond SPair-71k.

3.5.1. PF-WILLOW

PF-WILLOW [10] is derived from the WILLOW Object Class dataset and comprises 4 object categories — car, duck, motorbike, and wine bottle — with 10 keypoint annotations provided for each image. Each category is further partitioned into sub-classes based on viewpoint and background clutter conditions, for a total of 10 classes.

3.5.2. PF-PASCAL

The PF-PASCAL dataset [10] is built upon the PASCAL 2011 keypoint annotations and covers 20 object categories. Its test set consists of 299 image pairs. Overall, PF-PASCAL represents a more challenging benchmark than PF-WILLOW, while still being less demanding than SPair-71k in terms of scale, intra-class variation, viewpoint changes, and occlusion.

3.5.3. AP-10K

AP-10K [9] is a large-scale mammal pose estimation benchmark with 10,015 images from 23 families and 54 species, each with manually annotated keypoints. Thanks to these annotations, the dataset can also be used for semantic correspondence. To control the number of pairs and categories, we apply dedicated pre-processing. A fixed random seed ensures reproducibility, and any candidate pair with fewer than three shared keypoints is discarded. For intra-species pairs, we cap both the number of species and the number of pairs per species, then randomly sample valid pairs from all admissible intra-species pairs. For cross-species pairs, we impose analogous limits at the family level: only selected families are used, and for each valid family-level species combination, we randomly draw a fixed number of cross-species pairs. For cross-family pairs, we select a restricted set of families and uniformly sample a capped number of valid pairs for each family–family combination.

3.6. Extension: Diffusion-Based Correspondence Features

Diffusion models are predominantly employed for image synthesis conditioned on textual input. Stable Diffusion 2.1 is a text-to-image generative model in which the generation process is formulated as an iterative denoising procedure. The core architecture of SD 2.1 is a U-Net–based convolutional neural network with an encoder–decoder structure and extensive skip connections. These skip connections facilitate the integration of low-level spatial information from early layers with high-level semantic representations from deeper layers. At each diffusion timestep, the U-Net predicts the noise component to be subtracted from the latent representation, progressively refining the underlying image structure. Textual conditioning is provided by a pretrained OpenCLIP ViT-H/14 text encoder, which maps the input prompt to a sequence of semantic embeddings. These embeddings are injected into the U-Net via cross-attention layers.

Semantic correspondence arises emergently from the internal representations of diffusion models [7], even in the absence of explicit supervision for correspondence-related tasks. Earlier layers tend to encode more semantic and global features, whereas later layers primarily capture low-level, fine-grained details. In our setting, we extract DIFT features from upsampling blocks 0, 1, and 2 of the diffusion U-Net.

To reduce the variability caused by the stochastic noise injection in the diffusion process, each image is processed through an ensemble of noisy forward passes. In practice, the same input image is replicated along the batch dimension, and each copy is perturbed with an independently sampled noise realization. The feature maps extracted from these noisy versions are then averaged, producing a more stable and robust representation.

The timestep controls the progression of the diffusion process. For low timestep values, low-level image details are better preserved, whereas higher timestep values yield more abstract and semantic representations. In practice, effective timesteps typically lie in the range of approximately 100 to 300.

To further improve performance, it is possible to combine Stable Diffusion with DINOv2 [8] by stacking their feature map. Because the diffusion process operates by gradually denoising structured latent representations, Stable Diffusion is particularly effective at preserving local spatial relationships and densely organized structures across images. However, these latent representations are not explicitly optimized for semantic discriminability. As a result, visually similar regions that in fact belong to different semantic categories may be incorrectly associated, thereby limiting reliability in tasks that require strict semantic consistency.

In contrast, DINOv2 features are explicitly trained to maximize semantic alignment through large-scale self-supervised learning. This training paradigm makes them highly effective at identifying corresponding objects or object parts across different images, even in the presence of substantial variations in appearance. Nevertheless, this strong semantic robustness is achieved at the expense of spatial precision.

4. Experimental Results

This section reports the evaluation of the considered visual foundation models on SPair-71k. We use PCK at three thresholds, $\alpha = 0.05$, $\alpha = 0.10$, and $\alpha = 0.20$, reporting both PCK per point and PCK per image. The former averages over all keypoint correspondences, while the latter averages the accuracy over image pairs. Unless otherwise stated, results refer to the best-performing configuration for each setting; for light fine-tuning, we also report the best configuration for each number of fine-tuned layers.

4.1. Training-Free Baseline Results

Table 5 compares the zero-shot performance of DINOv2, DINOv3, and SAM on SPair-71k. DINOv2 provides the strongest training-free baseline, reaching 53.9 mean PCK per point at $\alpha = 0.10$ and 69.9 at $\alpha = 0.20$, with the same trend observed in the per-image scores (51.2 and 67.1). DINOv3 follows closely, obtaining 52.0 and 66.8 mean PCK per point at the same thresholds. Therefore, under our zero-shot protocol based on final-layer patch descriptors and nearest-neighbor matching, DINOv2 is slightly stronger overall. However, the gap is limited, and the category-wise results show that DINOv3 performs better on specific classes, such as Bottle, Cat, Plant, Train, and TV. This suggests that the two DINO backbones encode semantic and spatial information differently across categories, despite both relying on self-supervised visual representations. This behavior is consistent with previous work showing that DINO features can provide effective dense descriptors for correspondence, while still producing imperfect and sometimes sparse matching fields [6, 8].

Table 1. Overall SPair-71k evaluation for fine-tuned models with different numbers of fine-tuned layers.

Backbone	Layers	PCK per keypoint			PCK per image		
		$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$
DINOv2 ViT-B/14	1	**53.7**	**69.7**	**79.7**	**52.6**	**68.6**	**78.8**
	2	52.1	67.3	77.5	50.7	65.7	75.9
	3	50.68	66.22	76.35	48.92	64.10	74.29
DINOv3 ViT-B/16	1	59.7	75.4	85.0	58.0	73.6	83.5
	2	60.7	76.9	86.1	58.8	74.9	84.5
	3	**61.9**	**77.5**	**86.5**	**59.8**	**75.4**	**84.8**
SAM ViT-B/16	1	32.41	42.68	55.40	30.73	40.60	53.13
	2	33.13	43.48	56.20	31.43	41.39	53.91
	3	**33.60**	**44.0**	**56.62**	**31.88**	**41.81**	**54.14**

In contrast, SAM is substantially weaker in the same zero-shot setting, reaching only 23.1 mean PCK per point at $\alpha = 0.10$ and 36.1 at $\alpha = 0.20$, with per-image scores of 20.7 and 33.4. This large gap indicates that segmentation-oriented features do not directly transfer to fine-grained semantic keypoint matching. While SAM is designed for promptable mask prediction, SPair-71k evaluates the localization of corresponding semantic parts across object instances under strong viewpoint, scale, occlusion, and truncation variations [3]. Overall, the comparison shows that DINO-based self-supervised features are better suited to zero-shot semantic correspondence than SAM features: DINOv2 is the strongest baseline, DINOv3 remains competitive and shows category-specific strengths, while SAM requires further adaptation to become effective for part-level correspondence.

4.2. Effect of Light Fine-Tuning

Compared to the zero-shot baseline (Table 5), Table 6 demonstrates that our fine-tuning strategy yields significant improvements for DINOv2, DINOv3, and SAM on the SPair-71k dataset, successfully bridging the gap between high-level semantics and strict geometric localization.

We evaluated three configurations by unfreezing the last one, two, and three layers of the Vision Transformer. For DINOv2, unfreezing only the final layer gives the best overall performance. Deeper fine-tuning reduces the mean PCK, suggesting a trade-off between task adaptation and catastrophic forgetting. However, some atypical categories, such as TV and Plant, benefit from updating more layers, probably because their geometric or irregular structures require more specific feature adaptation.

DINOv3 achieves the best results across all evaluated models and benefits from deeper adaptation. Unlike DINOv2, it reaches its peak performance when the last three layers are unfrozen, suggesting a more robust representa-

Table 2. Effect of hard argmax and window soft-argmax on the fine-tuned models.

Backbone	Prediction	PCK per keypoint			PCK per image		
		$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$
DINOv2 ViT-B/14	Hard argmax	50.8	66.5	78.8	49.8	65.5	77.9
	Window soft-argmax	**53.7**	**69.7**	**79.7**	**52.6**	**68.6**	**78.8**
DINOv3 ViT-B/16	Hard argmax	59.3	75.2	85.6	57.2	73.3	83.9
	Window soft-argmax	**61.9**	**77.5**	**86.5**	**59.8**	**75.4**	**84.8**
SAM ViT-B/16	Hard argmax	32.3	42.8	56.3	30.7	40.7	53.8
	Window soft-argmax	**33.6**	**44.0**	**56.6**	**31.9**	**41.8**	**54.1**

tion space that can support stronger task-specific updates without severe forgetting.

Although the DINOv2 and DINOv3 architectures confirm their superiority in terms of absolute performance on the SPair-71k dataset, the comparative analysis highlights the exceptional impact of fine-tuning on SAM. Compared to its zero-shot baseline, SAM registers the most pronounced relative performance leap, achieving an increase of over 125% under stringent localization thresholds ($\alpha = 0.05$). This demonstrates remarkable plasticity within its ViT backbone: while the original pre-training penalizes fine-grained correspondences in favor of generating homogeneous masks, fine-tuning successfully realigns the internal representations, uncovering rich, latent semantic and spatial information. The validity of this realignment is further supported by accuracy peaks on morphologically structured categories (e.g. Bird 80.9%, Train 75.0% at $\alpha = 0.10$), proving SAM's competitiveness in dense semantic matching when appropriately supervised. Similar to DINOv3, the optimal performance is achieved by unfreezing the last three layers as shown in Table 1.

4.3. Effect of Window Soft-Argmax

Table 2 shows that window soft-argmax consistently improves PCK across all fine-tuned backbones. The gains are most evident at the stricter thresholds, especially for $\alpha = 0.05$ and $\alpha = 0.10$, while they become smaller at $\alpha = 0.20$. This indicates that the refinement mainly improves localization accuracy when the initial hard-argmax prediction is already close to the correct correspondence.

The improvement is substantial for both DINOv2 and DINOv3, with DINOv3 showing slightly larger gains at the strictest threshold, while the effect is more limited for SAM. This suggests that window soft-argmax is most effective when the backbone already produces a locally meaningful similarity response. In contrast, when the feature representation is weaker or the semantic match is incorrect, as more often observed with SAM, the refinement cannot fully compensate for the underlying matching error.

Overall, window soft-argmax should be interpreted as a lightweight inference-time refinement rather than a mechanism that changes the semantic quality of the features. It improves the precision of already plausible matches, but it does not solve failures caused by ambiguous or incorrect correspondences.

Table 3. Overall SPair-71k evaluation for LoRA models.

Backbone	PCK keypoint			PCK image		
	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$	$\alpha = 0.05$	$\alpha = 0.10$	$\alpha = 0.20$
DINOv2 LoRA	37.9	56.5	71.1	35.4	53.6	68.3
DINOv3 LoRA	56.2	71.9	81.8	54.0	69.3	79.5
SAM LoRA	29.3	40.7	54.4	27.2	37.9	51.3

4.4. Cross-Dataset Generalization Results

Table 7 reports a comparative evaluation of the DINOv2, DINOv3, and SAM backbones (in both zero-shot and fine-tuned configurations) on PF-WILLOW, PF-PASCAL, and AP-10K. The results show that finetuning is consistently beneficial: all models exhibit performance gains across all datasets. This outcome is expected for PF-WILLOW and PF-PASCAL, as these benchmarks are relatively simple and contain object categories that are closely related to those in SPair.

On AP-10K, a reduction in accuracy is observed when moving from the intra-species to the cross-family setting, reflecting the increased difficulty of matching keypoints across more dissimilar animal categories. Nevertheless, the accuracy drop remains limited for both zero-shot and fine-tuned models, indicating the robustness of both the backbone architectures and the finetuning procedure.

Overall, these experiments corroborate the trends observed on SPair-71k: DINOv2 and DINOv3 achieve strong performance in the zero-shot setting, whereas SAM underperforms relative to these models. After finetuning, DINOv3 surpasses DINOv2 by a substantial margin.

4.5. LoRA-Based Adaptation Results

Table 3 reports the results obtained with LoRA-based adaptation. LoRA improves all backbones with respect to their zero-shot baselines, but the final performance remains strongly dependent on the underlying representation. DINOv3 obtains the best LoRA results, reaching 71.9 mean PCK per keypoint at $\alpha = 0.10$ and 81.8 at $\alpha = 0.20$, with corresponding per-image scores of 69.3 and 79.5. DINOv2 also improves, but remains clearly below DINOv3, reaching 56.5 and 71.1 mean PCK per keypoint at the same thresholds. SAM benefits from LoRA as well, but obtains the weakest results, with 40.7 at $\alpha = 0.10$ and 54.4 at $\alpha = 0.20$.

This comparison shows that LoRA is effective as a lightweight adaptation strategy, but it does not affect all backbones equally. Since LoRA freezes the pretrained weights and learns low-rank updates, its adaptation capacity is more constrained than direct fine-tuning [5]. In our results, this constraint is less limiting for DINOv3, which remains the strongest model under LoRA adaptation, while DINOv2 shows a smaller gain and SAM remains far from

Table 4. Overall comparison between DIFT and DIFT + DINOv2 on SPair-71k.

Model	PCK@0.05	PCK@0.10	PCK@0.20
DIFT	43.25 / 39.33	54.89 / 50.49	64.76 / 60.47
DIFT + DINOv2	50.56 / 47.12	63.86 / 60.13	74.57 / 71.05

the DINO-based models. This suggests that parameter-efficient adaptation can improve semantic correspondence, but cannot fully compensate for the quality of the initial dense features. The trend is consistent with previous work showing that pretrained visual features can support correspondence, while their effectiveness depends on the semantic and spatial information encoded by the representation [7, 8].

4.6. Diffusion-Based Correspondence Results

Table 4 shows that diffusion features demonstrates very competitive performance, outperforming SAM and the zero-shot DINO backbones. This suggests that diffusion-based features are particularly effective for precise local correspondence, even without any task-specific adaptation. When compared with the fine-tuned backbones, the DINO-based models remain stronger, especially DINOv3. Nevertheless, diffusion features remains stronger than SAM and achieves competitive results. This indicates that diffusion features encode meaningful visual and semantic structures, although they are not as effective as task-adapted DINO descriptors for fine-grained keypoint matching.

These observations motivate the fusion between diffusion features and DINOv2. Diffusion features (DIFT) provides accurate diffusion-based local correspondences, while DINOv2 contributes more stable and semantically robust descriptors. Therefore, combining the two representations allows the method to exploit complementary information: DIFT improves local matching precision, whereas DINOv2 strengthens semantic reliability. As shown, this fusion consistently improves over DIFT alone across all PCK thresholds, confirming the benefit of integrating diffusion-based and self-supervised features. All tests were made with an ensemble size of 4 to ensure the maximum precision. However, empirical evidence shows that larger ensembles provide only marginal gains while increasing inference time and VRAM consumption.

5. Conclusion

In this work, we evaluated the use of visual foundation models for semantic correspondence on SPair-71k. We compared three different backbones, DINOv2, DINOv3, and SAM, under three adaptation regimes: a training-free baseline, light fine-tuning of the last layers, and LoRA-based adaptation. The results show that DINO-based self-supervised representations are substantially more suitable for semantic correspondence than SAM features, especially when used for fine-grained keypoint matching under large viewpoint and scale variations.

Among the evaluated models, DINOv3 achieves the best overall performance after light fine-tuning. In particular, fine-tuning the last three layers provides the strongest results, reaching the highest PCK scores across both per-keypoint and per-image metrics. DINOv2 remains a strong baseline, especially in the zero-shot setting, but its performance decreases when too many layers are unfrozen, suggesting a higher sensitivity to overfitting. SAM remains the weakest model overall—confirming that segmentation-oriented representations are not directly optimized for semantic part correspondence—but it exhibits the greatest relative improvement after fine-tuning.

We also analyzed the effect of the prediction rule. Window soft-argmax consistently improves over hard argmax, especially at stricter thresholds, where small localization errors have a stronger impact. This confirms that local refinement is useful when the similarity map already identifies the correct semantic region, but it cannot compensate for incorrect semantic matches. Finally, LoRA provides an efficient adaptation strategy and improves all backbones with respect to their zero-shot baselines. However, it does not match the best light fine-tuning results, indicating that for this task a small amount of direct backbone adaptation is more effective than low-rank updates alone.

Finally, we investigated diffusion-based features. The results show that those are highly competitive in the training-free setting, outperforming SAM and the zero-shot DINO backbones at stricter thresholds. The fusion between DIFT and DINOv2 further improves the results, confirming that diffusion-based and self-supervised descriptors provide complementary information.

Overall, our results indicate that the best trade-off between accuracy and adaptation cost is obtained by lightly fine-tuning DINOv3 and applying window soft-argmax at inference time. Future work could further investigate hybrid representations, such as combining DINO features with diffusion-based features, since prior work suggests that diffusion models provide complementary spatial information for dense correspondence [7, 8]. Additional experiments on PF-WILLOW, PF-PASCAL, and AP-10K could also provide a broader evaluation of cross-dataset generalization.

Table 5. SPair-71k zero-shot evaluation comparison among DINOv2 ViT-B/14, DINOv3 ViT-B/16, and SAM ViT-B/16. The top section reports PCK per point results, while the bottom section reports PCK per image results.

Method	α	**SPair-71k Category**																		
		Aero	Bike	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Dog	Horse	Motor	Person	Plant	Sheep	Train	TV	**Mean**
PCK per point results																				
	0.05	54.7	40.8	64.6	19.4	24.1	35.1	31.3	57.7	25.4	46.5	43.9	42.9	38.2	45.5	16.0	40.4	30.8	13.7	36.9
DINOv2 (Zero-shot)	0.10	69.6	59.9	84.2	33.4	41.4	50.2	47.5	67.8	36.3	66.4	65.1	66.4	56.4	66.0	31.5	60.4	49.3	27.4	53.9
	0.20	82.2	75.0	94.8	52.7	59.6	61.8	59.2	80.9	52.2	82.0	79.8	82.4	74.0	79.7	54.8	77.2	70.6	43.7	69.9
	0.05	39.6	33.8	53.4	13.9	30.0	33.5	31.9	58.6	24.6	45.5	37.2	37.0	25.7	40.7	22.9	32.6	36.0	29.1	34.8
DINOv3 (Zero-shot)	0.10	54.0	49.8	75.5	25.5	45.2	47.8	47.5	69.3	37.3	65.2	54.3	56.5	43.2	61.3	44.1	52.2	55.2	51.8	52.0
	0.20	67.2	64.2	87.9	40.5	59.7	58.9	58.7	80.1	51.9	79.9	70.7	72.5	60.7	75.5	63.2	68.9	72.5	70.0	66.8
	0.05	18.0	10.2	20.3	10.5	17.3	13.5	15.0	28.3	9.4	19.1	12.9	9.9	8.9	18.2	13.1	9.1	19.7	14.3	14.9
SAM (Zero-shot)	0.10	26.0	16.2	30.9	18.2	26.8	18.1	20.6	39.4	13.9	27.3	20.8	16.4	16.4	30.7	23.1	15.6	30.5	24.6	23.1
	0.20	39.9	27.4	45.2	33.0	41.9	28.1	30.7	55.0	22.5	40.8	35.4	27.9	26.9	45.3	33.4	26.8	49.9	40.5	36.1
PCK per image results																				
	0.05	53.0	37.6	65.1	19.5	23.3	28.9	24.5	57.9	23.8	44.1	42.2	41.1	36.7	42.2	15.4	37.4	28.6	12.6	34.6
DINOv2 (Zero-shot)	0.10	68.6	56.5	83.9	33.0	40.7	41.4	39.0	68.2	33.9	64.9	63.6	64.3	53.9	62.6	30.1	55.9	47.2	25.5	51.2
	0.20	81.5	71.1	94.9	51.1	59.2	51.6	51.4	81.3	49.6	81.5	79.1	80.4	71.0	76.2	53.0	72.7	69.5	40.6	67.1
	0.05	36.5	31.3	51.3	12.1	28.0	25.0	23.0	58.3	22.1	41.7	35.1	35.6	24.6	37.6	20.1	28.5	33.0	26.3	31.7
DINOv3 (Zero-shot)	0.10	51.3	46.4	73.5	22.1	44.3	36.0	34.5	69.4	34.4	60.9	52.3	54.4	41.5	57.0	40.9	46.5	52.6	46.4	48.0
	0.20	64.5	60.2	86.1	35.1	59.1	45.7	45.6	80.3	48.4	76.4	69.4	69.9	58.3	71.1	59.6	62.7	69.4	63.1	62.5
	0.05	15.9	9.0	19.4	8.4	16.2	10.6	10.1	28.4	8.1	15.8	11.9	8.9	8.4	15.3	10.9	7.2	17.7	13.9	13.1
SAM (Zero-shot)	0.10	23.6	14.5	29.7	14.7	25.5	14.3	14.6	39.5	12.3	23.8	19.3	14.9	16.0	26.9	19.6	12.4	28.0	23.8	20.7
	0.20	37.2	24.9	43.6	27.9	40.5	23.1	24.6	55.2	20.3	35.9	33.6	26.3	26.6	42.0	30.4	22.9	47.0	39.5	33.4

Table 6. SPair-71k evaluation comparison among fine-tuned DINOv2 ViT-B/14, fine-tuned DINOv3 ViT-B/16, and fine-tuned SAM ViT-B/16. The top section reports PCK per point results, while the bottom section reports PCK per image results.

Method	α	**SPair-71k Category**																		
		Aero	Bike	Bird	Boat	Bottle	Bus	Car	Cat	Chair	Cow	Dog	Horse	Motor	Person	Plant	Sheep	Train	TV	**Mean**
PCK per point results																				
	0.05	70.7	56.8	76.9	40.6	35.2	61.1	59.4	68.5	43.3	57.1	59.6	57.4	64.7	62.2	30.0	45.4	47.6	40.9	53.7
DINOv2 (fine-tuned)	0.10	79.3	70.2	91.0	59.8	51.7	73.2	72.0	76.6	50.1	77.0	78.9	80.0	78.5	79.1	50.3	67.0	67.3	59.2	69.7
	0.20	86.7	78.7	96.9	76.3	64.8	78.3	78.4	86.1	57.4	86.6	88.8	91.5	87.5	88.2	68.9	79.4	78.1	68.5	79.7
	0.05	66.7	52.8	78.2	44.9	48.4	71.2	68.4	81.5	45.3	72.4	66.5	60.1	52.5	58.9	53.8	54.7	72.9	64.5	61.9
DINOv3 (fine-tuned)	0.10	77.2	68.0	91.1	58.8	65.8	82.6	78.8	87.3	56.5	89.0	81.5	80.9	70.8	76.5	77.2	73.2	89.5	90.6	77.5
	0.20	85.1	78.2	96.3	74.9	76.4	86.8	84.8	92.7	66.6	94.4	90.5	90.8	82.6	86.8	88.9	85.1	96.8	98.5	86.5
	0.05	36.6	21.9	50.3	22.6	28.1	37.0	32.5	52.2	23.3	39.6	32.9	30.4	23.3	37.3	31.0	26.5	45.3	34.0	33.6
SAM (fine-tuned)	0.10	44.9	29.2	67.1	30.1	37.7	41.9	38.3	61.1	26.4	49.8	44.2	43.1	32.4	51.9	45.5	39.1	58.9	50.4	44.0
	0.20	58.6	39.9	80.9	45.9	51.0	49.0	48.5	72.5	32.9	65.2	58.0	57.1	43.7	64.5	56.5	52.9	75.0	67.4	56.6
PCK per image results																				
	0.05	72.4	55.5	77.7	44.0	34.2	59.9	54.7	68.7	43.3	55.7	57.8	56.8	64.1	61.6	28.8	42.2	44.8	39.8	52.6
DINOv2 (fine-tuned)	0.10	80.8	68.8	91.9	62.8	50.8	71.3	67.5	76.8	51.1	76.7	77.9	78.8	77.6	78.5	48.5	62.1	65.9	58.9	68.6
	0.20	87.5	76.7	97.1	78.4	63.6	76.8	74.4	86.3	58.6	86.9	88.2	89.9	86.6	87.8	67.2	75.2	77.7	68.9	78.8
	0.05	67.9	50.2	77.5	48.3	46.6	67.0	61.8	81.7	42.8	69.4	63.3	58.7	50.8	56.6	49.5	49.7	71.1	62.7	59.8
DINOv3 (fine-tuned)	0.10	78.0	64.8	90.6	61.7	64.7	77.5	71.8	87.5	55.0	86.8	79.1	79.7	69.5	74.4	73.2	66.2	88.7	88.2	75.4
	0.20	85.2	74.9	96.0	76.8	76.0	81.7	79.7	92.9	65.6	93.1	89.2	89.8	81.5	85.9	85.5	78.3	95.8	98.0	84.8
	0.05	35.3	21.4	50.5	24.9	26.2	33.7	27.2	53.0	21.3	37.6	30.8	29.0	22.7	34.5	26.6	23.7	43.1	32.4	31.9
SAM (fine-tuned)	0.10	43.2	28.2	67.4	31.6	36.1	38.3	32.3	61.8	24.0	48.4	41.7	41.5	31.9	48.8	38.6	33.7	57.2	47.9	41.8
	0.20	56.7	38.6	81.1	45.9	49.5	45.4	43.3	73.2	30.0	62.7	55.6	54.9	42.9	62.1	48.8	46.1	73.1	64.8	54.1

Table 7. Comparison of standard and finetuned models on other datasets. In PF-WILLOW, the PCK per keypoint and the PCK per image coincide because each image pair contains the same number of annotated keypoints. Therefore, the image-level aggregation does not introduce a different weighting with respect to the keypoint-level aggregation, and both metrics produce the same values.

Method	**PF-WILLOW**			**PF-PASCAL**			**AP-10K INTRA-SP**			**AP-10K INTRA-FM**			**AP-10K CROSS-FM**		
	PCK@0.05	PCK@0.10	PCK@0.20	PCK@0.05	PCK@0.10	PCK@0.20	PCK@0.05	PCK@0.10	PCK@0.20	PCK@0.05	PCK@0.10	PCK@0.20	PCK@0.05	PCK@0.10	PCK@0.20
DINOv2	31.6	58.9	85.2	48.2 / 48.4	68.4 / 52.8	87.7 / 67.8	35.3 / 36.91	54.5 / 55.2	73.3 / 73.9	33.0 / 34.8	53.0 / 53.9	73.4 / 73.6	29.4 / 31.6	45.9 / 47.5	62.9 / 63.3
DINOv2 (our)	42.0	68.0	86.9	55.7 / 55.4	72.7 / 72.5	84.4 / 84.1	39.8 / 41.3	59.7 / 59.7	76.7 / 75.0	35.9 / 37.7	56.8 / 57.6	74.4 / 73.5	31.8 / 33.9	48.1 / 49.8	62.8 / 63.0
SAM	22.2	36.1	51.9	24.5 / 24.6	34.3 / 34.3	48.8 / 48.9	9.6 / 10.0	18.6 / 18.5	34.1 / 33.7	8.1 / 8.4	16.2 / 16.2	33.4 / 33.2	6.0 / 6.3	11.9 / 11.9	25.1 / 24.6
SAM (our)	29.1	43.5	58.2	31.2 / 31.3	43.2 / 43.3	58.1 / 58.0	20.7 / 21.6	30.4 / 30.5	45.6 / 45.5	16.7 / 17.3	26.1 / 26.8	43.4 / 44.3	13.2 / 14.0	22.1 / 22.5	37.0 / 37.0
DINOv3	36.7	65.9	88.9	50.0 / 50.1	74.8 / 75.0	91.0 / 91.0	32.8 / 34.6	51.5 / 53.0	69.2 / 69.6	29.5 / 31.2	49.2 / 50.3	68.9 / 69.3	25.7 / 28.0	40.1 / 42.3	56.5 / 57.8
DINOv3 (our)	47.9	75.0	92.2	61.4 / 61.3	78.0 / 77.8	90.5 / 90.0	46.3 / 49.4	64.4 / 66.0	79.0 / 78.6	42.5 / 46.3	61.9 / 63.8	77.8 / 77.7	40.0 / 43.8	55.3 / 57.7	67.7 / 69.0

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